

L A T E X M A C R O S & N O T A T I O N

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NOTATION SUMMARY

Scalar values are represented by lowercase italic letters (`\mathit`), e.g., x . Column vectors are represented by kets (`\ket`), e.g., $|x\rangle$, and row vectors by bras (`\bra`), e.g., $\langle x|$. Matrices/operators are represented by uppercase italic letters (`\mathit`), e.g., X . Functions and functionals are represented by computer modern letters (`\mathrm`), e.g., $x(\cdot)$ or $X(\cdot)$. Superoperators are represented by uppercase calligraphic letters (`\mathcal`), e.g., $\mathcal{X}(\cdot)$. Random variables are represented by uppercase sans serif letters (`\mathsf`), e.g., X . Sets of elements are represented by uppercase blackboard bold letters (`\mathbbol`), e.g., $\mathbb{X} \equiv \{x_1, x_2, \dots, x_n\}$. Tuples are represented by capital typewriter typestyle letters (`\mathtt`), e.g., X . Mathematical spaces are represented by uppercase Ralph Smith formal script letters (`\mathscript`), e.g., $\mathscr{X}$. Mathematical transformations are represented by uppercase Euler script letters (`\EuScript`), e.g., $\mathcal{X}$. Probability and information theory related concepts are represented by uppercase blackboard bold letters (`\mathbb`), e.g., $\mathbb{X}$. Mathematical groups are represented by uppercase AMS bold letters (`\mathbf`), e.g., $\mathbf{X}$. The complex conjugate of a scalar x is denoted as x^* , the transpose of a matrix X as $X^\top$, and the conjugate transpose of a matrix X as $X^\dagger$. The imaginary unit is denoted as $\imath := \sqrt{-1}$, and the identity ($d \times d$)-dimensional matrix as $\mathbb{1}_d$.

MATHEMATICAL FONTS

```
\usepackage{physics}% Physics package including Dirac notation
\usepackage{amsmath}% Math package for miscellaneous enhancements
\usepackage{amssymb}% Package for mathbb font
\usepackage{dsfont}% Package for mathds font
\usepackage{mathrsfs}% Calligraphic math letters
\usepackage{euscript}% Calligraphic math letters
\usepackage{upgreek}% Package for alternative greek letters
\usepackage{mathtools}% Package for extra mathematical symbols
\usepackage{mathdots}% Package for making dot circles
\usepackage[bbgreekl]{mathbbol}% Package for mathbbol font
\DeclareSymbolFontAlphabet{\mathbbol}{bbold}% Declares command \mathbbol
\DeclareSymbolFontAlphabet{\mathbb}{AMSb}% Declares command \mathbb
```

MATHEMATICAL COMMANDS AND CONVENTIONS

```

\newcommand{\m}[1]{\mathit{#1}}% Matrix/Operator
\newcommand{\f}[1]{\mathrm{#1}}% Function
\newcommand{\map}[1]{\ifcat\noexpand#1\relax#1\else{\mathcal{#1}}\fi}% Superoperator
\newcommand{\set}[1]{\mathbbol{#1}}% Set
\newcommand{\tuple}[1]{\mathhtt{#1}}% Tuple
\newcommand{\rnd}[1]{\mathsf{#1}}% Random variable
\newcommand{\group}[1]{\mathbf{#1}}% Mathematical group
\newcommand{\est}[1]{\hat{#1}}% Estimated parameter
\newcommand{\arccosh}{\mathrm{arccosh}\hspace{0.04cm}}
\newcommand{\arcsinh}{\mathrm{arcsinh}\hspace{0.04cm}}
\newcommand{\arctanh}{\mathrm{arctanh}\hspace{0.04cm}}
\newcommand{\arccoth}{\mathrm{arccoth}\hspace{0.04cm}}
\newcommand{\tra}{\intercal}% Replace \intercal with \tra
\newcommand{\diagarrow}{\mathrel{\text{$\nearrow$}\llap{$\searrow$}}}% Diagonal double arrow
\newcommand{\adiagarrow}{\mathrel{\text{$\nwarrow$}\llap{$\searrow$}}}% Anti-diagonal double arrow
\newcommand{\braf}[1]{\mbox{$\angle$ #1 |$}}% Bra with fixed height
\newcommand{\ketf}[1]{\mbox{$|$ #1 \angle$}}% Ket with fixed height
\newcommand{\braketf}[2]{\mbox{$\angle$ #1 | #2 \angle$}}% Braket with fixed height
\newcommand{\ketbraf}[2]{\mbox{$\vert$ #1 \angle$\!\!\angle$ #2 \vert$}}% Ketbra with fixed height
\newcommand{\ma}[1]{\begin{bmatrix} #1 \end{bmatrix}}% Shortcut for matrix

```

```

\newcommand{\eq}[2]{\begin{equation} \label{eq:#1} #2 \end{equation}}% Shortcut for equation
\newcommand{\al}[2]{\begin{align} \label{eq:#1} #2 \end{align}}% Shortcut for align
\newcommand{\alsub}[2]{\begin{subequations}\label{eq:#1}\begin{align} #2 \end{align}\end{subequations}}
% Shortcut for align sub-equations
\newcommand{\ga}[2]{\begin{gather} \label{eq:#1} #2 \end{gather}}% Shortcut for gather
\newcommand{\gasub}[2]{\begin{subequations}\label{eq:#1}\begin{gather} #2 \end{gather}\end{subequations}}
% Shortcut for gather sub-equations
\let\leq\leqslant% Replace \leq with \leqslant
\let\geq\geqslant% Replace \geq with \geqslant

```

- **Scalar**

$x \rightarrow x$ or $\lambda \rightarrow \lambda$

- **Vector**

$\ket{x} \rightarrow |x\rangle$ or $\ket{\psi} \rightarrow |\psi\rangle$
 $\vec{x} \rightarrow \vec{x}$ or $\vec{\psi} \rightarrow \vec{\psi}$

- **Tuple**

$\tuple{X} \rightarrow X$

- **Matrix/Operator**

$\mathsf{M}\{X\} \rightarrow X$ or $\mathsf{M}\{\Psi\} \rightarrow \Psi$

- **Function/Functional**

$f\{f\}(x) \rightarrow f(x)$ or $f\{f\}(\lambda) \rightarrow f(\lambda)$
 $f\{f\}(\mathsf{M}\{X\}) \rightarrow f(X)$ or $f\{f\}(\Lambda) \rightarrow f(\Lambda)$

- **Superoperator**

$\mathsf{map}\{E\}(\mathsf{M}\{X\}) \rightarrow \mathcal{E}(X)$
 $\mathsf{map}\{\mathsf{Uplambda}\}(X) \rightarrow \Lambda(X)$

- **Mathematical set**

$\mathsf{set}\{X\} \rightarrow \mathbb{X}$ or $\mathsf{set}\{\Omega\} \rightarrow \Omega$
Alternatively a set is denoted as $\mathbb{X} \equiv \{x_1, \dots, x_n\}$

- **Mathematical tuple**

$\tuple{X} \rightarrow X$

- **Mathematical space**

$\mathsf{mathscr}\{X\} \rightarrow \mathcal{X}$

- **Random variable**

$\mathsf{rnd}\{X\} \rightarrow X$

- **Probability and information theory expression**

$\mathsf{mathbb}\{X\} \rightarrow \mathbb{X}$

- **Mathematical group**

$\mathsf{group}\{X\} \rightarrow \mathbf{X}$

- **Estimated parameter**

$\mathsf{est}\{x\} \rightarrow \hat{x}$

- **Matrix**

$\mathsf{ma}\{[a \ \& \ b \ \& \ c \ \& \ d]\} \rightarrow \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

- **Plain equation**

$\mathsf{eq}\{[label]\}\{[equation]\}$.

- **Aligned equation**

$\mathsf{al}\{[label]\}\{[equation]\}$.

- **Aligned within sub-equations format**

$\mathsf{alsub}\{[label]\}\{[equation]\}$.

- **Gathered equation**
 $\backslash\mathrm{ga}\{[label]\}\{[equation]\}.$
- **Gathered within sub-equations format**
 $\backslash\mathrm{gasub}\{[label]\}\{[equation]\}.$

SYMBOL SHORTCUTS

```

\newcommand{\qs}{S}% Generic quantum state
\newcommand{\id}[1]{\mathds{1}_{\{#1\}}}% Identity matrix
\newcommand{\hil}{\mathscr{H}}% Hilbert space
\newcommand{\ban}{\mathscr{B}}% Banach space
\newcommand{\leb}{\mathscr{L}^2}% Lebesgue square space
\newcommand{\eure}{\mathscr{R}}% Euclidean space over real numbers
\newcommand{\euco}{\mathscr{C}}% Euclidean space over complex numbers
\newcommand{\trc}{\mathrm{tr}}% Trace
\newcommand{\ptrc}[1]{\mathrm{tr}_{\{#1\}}}% Partial Trace
\newcommand{\com}[2]{\left[#1,#2\right]_{-}}% Commutator
\newcommand{\acom}[2]{\left[#1,#2\right]_{+}}% Anti-commutator
\newcommand{\prob}{\mathbb{P}}% Probability
\newcommand{\expv}{\mathbb{E}}% Expected Value
\newcommand{\vari}{\mathbb{V}}% Variance
\newcommand{\std}{\upsigma}% Standard deviation
\newcommand{\cov}{\mathbb{C}\mathrm{ov}}% Covariance
\newcommand{\cor}{\mathbb{C}\mathrm{or}}% Correlation
\newcommand{\pcc}{\mathbb{R}}% Total correlations
\newcommand{\ent}{\mathbb{H}}% Entropy
\newcommand{\mi}{\mathbb{I}}% Mutual information
\newcommand{\four}{\mathrm{EuScript{F}}}% Fourier transform
\newcommand{\su}{\mathrm{group{SU}}}% Special unitary group
\newcommand{\card}{\mathrm{card}}% Cardinality
\newcommand{\onorm}[1]{\left\|\mathrm{left\kern-0.25ex\left\kern-0.25ex\left\kern-0.25ex\left\kern-0.25ex\right\kern-0.25ex\right\kern-0.25ex\right\kern-0.25ex\right\kern-0.25ex\right\kern-0.25ex\right\}} % Operator norm

```

- **Quantum state:** S , generated by $\backslash\mathrm{qs}$
- **Identity matrix:** $\mathbb{1}_2$, generated by $\backslash\mathrm{id}\{2\}$
- **Hilbert space:** $\mathcal{H}$, generated by $\backslash\mathrm{hil}$
- **Banach space:** $\mathcal{B}$, generated by $\backslash\mathrm{ban}$
- **Lebesgue square space:** $\mathcal{L}^2$, generated by $\backslash\mathrm{leb}$
- **Euclidean space over real numbers:** $\mathcal{R}$, generated by $\backslash\mathrm{eure}$
- **Euclidean space over complex numbers:** $\mathcal{C}$, generated by $\backslash\mathrm{euco}$
- **Trace:** $\mathrm{tr}X$, generated by $\backslash\mathrm{trc} X$
- **Partial trace:** $\mathrm{tr}_A X$, generated by $\backslash\mathrm{ptrc}\{A\} X$
- **Commutator** is denoted as $[X, Y]_-$, generated by $\backslash\mathrm{com}\{X\}\{Y\}$
- **Anti-commutator** is denoted as $[X, Y]_+$, generated by $\backslash\mathrm{acom}\{X\}\{Y\}$
- **Probability:** $\mathbb{P}(X)$, generated by $\backslash\mathrm{prob}(X)$
- **Expectation value:** $\mathbb{E}(X)$, generated by $\backslash\mathrm{expv}(X)$
- **Variance:** $\mathbb{V}(X)$, generated by $\backslash\mathrm{vari}(X)$
- **Standard deviation:** σ_X , generated by $\backslash\mathrm{std}_X$
- **Covariance:** $\mathbb{Cov}(X, Y)$, generated by $\backslash\mathrm{cov}(X, Y)$

- **Pearson correlation coefficient:** $\mathbb{R}(X, Y)$, generated by `\pcc(X, Y)`
- **Entropy:** $\mathbb{H}(X)$, generated by `\ent(X)`
- **Mutual information:** $\mathbb{I}(X; Y)$, generated by `\mi(X; Y)`
- **Fourier transform:** $\mathcal{F}$, generated by `\four`
- **Special unitary group:** SU , generated by `\su`
- **Cardinality of a set:** card , generated by `\card`
- **Operator norm:** $\|X\|$, generated by `\onorm{X}`